Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 00 (2018) 000–000

Procedia Computer Science

www.elsevier.com/locate/procedia

The 17th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN 2026), October 28–30, 2026, Almaty, Kazakhstan

Evaluating an EMA-Adaptive LSTM for Continuous Behavioral Authentication on Mobile Devices

Madi Gali[a,*], Vassiliy Serbin[b], Aray Kassenkhan[c]

[a] *Department of Cybersecurity, Institute of Automation and Information Technologies, Satbayev University, Almaty 050013, Kazakhstan*
[b] *Department of Information Systems, Institute of Automation and Information Technologies, Satbayev University, Almaty 050013, Kazakhstan*
[c] *Department of Software Engineering, Institute of Automation and Information Technologies, Satbayev University, Almaty 050013, Kazakhstan*

Abstract

Continuous authentication from smartphone inertial sensors must remain useful when a legitimate user's score distribution changes, yet an adaptive decision rule can also increase false acceptance or become vulnerable to profile poisoning. We evaluate a two-layer long short-term memory (LSTM) classifier and a lightweight agent that updates the first two exponential moving moments of accepted confidence scores. The study uses accelerometer and gyroscope data from 100 HMOG subjects. To prevent leakage, the revised protocol reads only the physical sensor columns, constructs non-overlapping sequences within individual sessions, keeps complete sessions disjoint between model fitting, validation, and test, and estimates normalization and initial operating points without test data. Linear SVM, random forest, and LSTM receive the same 10-second input. On clean held-out sessions, their macro equal error rates are respectively 29.92% ± 0.80%, 24.05% ± 0.86%, and 21.87% ± 1.01%. At the enrollment-derived operating point, the frozen LSTM threshold and the EMA agent obtain HTERs of 23.19% ± 0.95% and 24.53% ± 1.28% in a balanced alternating zero-effort attack stress test. Under a standardized synthetic drift of $\|\delta\|_2 = 2$, their HTERs are 24.89% ± 1.37% and 26.39% ± 2.12%. Thus the tested EMA rule does not improve the frozen threshold. These results provide a direct ablation of the adaptive component and delimit, rather than assume, its benefit. The work's primary contribution is a reproducible, session-disjoint evaluation of online score adaptation and an explicit analysis of its security–usability trade-off.

© 2026 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)
Peer-review under responsibility of the Conference Program Chairs.

Keywords: continuous authentication; behavioral biometrics; LSTM; adaptive threshold; inertial sensors; HMOG; concept drift

1. Introduction

One-time authentication cannot protect an unlocked smartphone after control of the device changes. Continuous authentication (CA) instead evaluates behavioral signals throughout a session, often without explicit interaction [1, 2].

* Corresponding author.
E-mail address: Gali.M@stud.satbayev.university (Aray Kassenkhan).

1877-0509 © 2026 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)
Peer-review under responsibility of the Conference Program Chairs.

Accelerometers and gyroscopes are attractive for this purpose because they are widely available and capture user-specific handling and motion patterns [3, 4].

The difficulty is temporal change. A classifier trained during enrollment can become poorly calibrated as posture, task, fatigue, or context changes [5]. Retraining a deep model after every observation is impractical on a mobile device, whereas a score-level update is inexpensive. That convenience is not evidence of effectiveness: an update rule must improve false rejection without opening the operating point to impostors, and it must be evaluated causally rather than by selecting a second threshold on test data.

Earlier versions of this study did not isolate that claim. In particular, an EMA profile was defined but did not participate in the reported decision equation, LSTM-only ablation and drift experiments were absent, and the data split was insufficiently documented. The revised study addresses those issues directly. Its contributions are:

- a session-disjoint nested protocol in which raw windows and overlapping contexts cannot cross fit, validation, or test partitions;
- a decision equation in which the EMA mean and variance determine the online threshold, with parameters fixed before test evaluation;
- a controlled LSTM-only versus LSTM+agent ablation at the same initial operating point, reporting causal FAR, FRR, and HTER decisions; and
- clean-session and synthetic-drift results with explicit limitations on external validity and on the interpretation of synthetic covariate shift.

2. Related Work and Scope of Novelty

CA has a substantial literature. Broad surveys cover system components, modalities, metrics, privacy, and open deployment problems [6, 7, 2]. A contemporary smartphone-focused survey catalogs more than 140 sensor and behavioral-biometric approaches [4]. The present work therefore does not claim that continuous authentication, inertial sensing, LSTM classification, or EMA by itself is new.

HMOG introduced hand-movement, orientation, and grasp features for smartphone authentication [3]. Subsequent work applied augmentation, deep sensor models, and adaptive verification to mobile behavioral data [8, 9, 5]. LSTM and hybrid recurrent networks are well established for sequential inertial biometrics [10, 11, 12]. Exponentially weighted statistics are likewise standard tools for tracking non-stationary streams [13, 14].

Table 1. Positioning of the present evaluation. A check mark means the cited work explicitly contains the listed element; protocols and datasets differ, so reported error rates are not directly comparable.

Work	Inertial	Sequence model	Online update	Drift test
Sitová et al. [3]	✓	–	–	–
AUToSen [9]	✓	✓	–	–
Verma et al. [5]	✓	–	✓	✓
This study	✓	✓	✓	✓

The narrower novelty is the controlled combination of these elements: a session-disjoint comparison in which a frozen LSTM threshold and an EMA-updated threshold start from the same enrollment profile and face the same chronological test scores. This design isolates adaptation rather than attributing the full classifier performance to an "intelligent agent."

3. Dataset and Leakage-Resistant Protocol

3.1. HMOG data and parsing

HMOG contains 24 smartphone-interaction sessions for each of 100 participants, including sitting and walking tasks [3]. We use the accelerometer and gyroscope. An HMOG inertial CSV row contains metadata before the three

physical axes; consequently, only columns 4–6 are read as x, y, z. This detail is security-critical because a session identifier must never enter the feature vector. Six empty sensor files belonging to one subject are logged and skipped; that subject retains enough valid sessions to appear in every partition.

After a five-second offset, the first 20 non-overlapping windows of 100 samples are retained from each valid session. Each window is represented by mean, standard deviation, minimum, maximum, root mean square, and skewness for each of six channels, producing 36 features. Ten adjacent windows form a 10-second sequence. The sequence step is also ten windows, so evaluation endpoints do not share raw samples. Sequences never cross a session boundary.

3.2. Nested session split

For session number r, define $g(r) = (r-1) \bmod 5$. In outer fold f, sessions with $g(r) = f$ are test sessions, sessions with $g(r) = (f+1) \bmod 5$ are validation/enrollment sessions, and the remaining sessions are used to fit the model. Whole sessions, rather than sequences, are assigned to partitions. Every included subject is required to occur in all three partitions. This is a session-disjoint cross-validation protocol, not a claim of leave-one-subject-out or prospective unseen-subject recognition.

A standard scaler is fitted only to the model-fitting sequences and then applied unchanged to validation and test sequences. Early stopping and initial threshold estimation use validation data; test labels are not used for either.

4. Method

4.1. System overview

The system has five stages:

$$\boxed{\text{sensor samples}} \rightarrow \boxed{\text{1-s windows}} \rightarrow \boxed{\text{36 features}} \rightarrow \boxed{\text{LSTM score}} \rightarrow \boxed{\text{static or EMA decision}}$$

The first four stages are identical in the ablation. Only the last-stage threshold update changes, which makes any difference in FAR/FRR attributable to the agent rather than to a different classifier or observation window.

4.2. LSTM classifier and baselines

The input is $\mathbf{X} \in \mathbb{R}^{10\times 36}$. A two-layer LSTM with 64 hidden units per layer returns its last hidden state. Layer normalization, a 32-unit ReLU layer, dropout of 0.3, and a 100-class softmax produce claimant scores $s_{u,n} \in [0, 1]$. Training uses cross-entropy, Adam with learning rate 10^{-3}, batch size 256, a maximum of 20 epochs, and early-stopping patience 4.

Linear SVM and random forest baselines receive the same sequence flattened to 720 dimensions. The forest contains 300 trees and has minimum leaf size 2. These baselines compare classifier families at the same temporal endpoint; they are not presented as adaptive systems.

4.3. EMA adaptive agent

For claimant u, the validation genuine scores initialize the first and second moments $p_{u,0}$ and $q_{u,0}$. Before attempt n, the agent computes

$$\sigma_{u,n-1} = \sqrt{\max(q_{u,n-1} - p_{u,n-1}^2, \epsilon^2)}, \tag{1}$$

$$\tau_{u,n} = \max(0, p_{u,n-1} - k\sigma_{u,n-1}), \tag{2}$$

$$d_{u,n} = \mathbb{I}[s_{u,n} \geq \tau_{u,n}]. \tag{3}$$

When $d_{u,n} = 1$, the accepted score updates both moments:

$$p_{u,n} = (1 - \alpha)p_{u,n-1} + \alpha s_{u,n}, \tag{4}$$

$$q_{u,n} = (1 - \alpha)q_{u,n-1} + \alpha s_{u,n}^2. \tag{5}$$

After a rejection the state is unchanged. Thus repeated unauthenticated events cannot lower the threshold. An accepted impostor can still poison the profile; the evaluation deliberately preserves that risk instead of using ground-truth labels to gate updates.

We fix $\alpha = 0.05$, corresponding to an effective memory of approximately $1/\alpha = 20$ accepted attempts. For each claimant, the validation genuine and impostor scores determine an EER threshold $\tau_{u,0}$; the corresponding $k_u = (p_{u,0} - \tau_{u,0})/\sigma_{u,0}$ makes the first adaptive threshold exactly equal to that independently calibrated operating point. Thus k_u is not selected on test scores. The frozen LSTM ablation uses the same $\tau_{u,0}$ but never updates it. There is no separate sliding mean and no unexplained $K = 10$ score buffer.

5. Evaluation

For each claimant, equal error rate (EER) is found from all exact ROC thresholds on held-out scores, then macro-averaged across claimants and folds. This EER is a score-separability statistic; its test-derived crossing threshold is not used for online decisions.

The static and adaptive decisions are evaluated at their enrollment-derived operating points. Genuine test attempts follow session and within-session chronology. Each is alternated with a zero-effort impostor score sampled without replacement; sampling is repeated ten times. This balanced 50% attack stream is an explicit stress test, not an estimate of real attack prevalence. Because the agent sees no labels, an accepted impostor updates its profile. We report macro false acceptance rate (FAR), false rejection rate (FRR), and HTER = (FAR + FRR)/2.

To isolate score drift, a deterministic direction in standardized 36-feature space is added progressively over each subject's test chronology. Magnitudes $\|\delta\|_2 \in \{0, 1, 2\}$ are tested on the same trained model. This is a controlled covariate-shift experiment; it does not claim to reproduce every physiological form of natural behavioral drift.

6. Results

Table 2. Clean held-out-session macro EER (mean ± standard deviation over five outer folds). All methods use the same 10-second sequences.

Model	EER (%)
Linear SVM	29.92 ± 0.80
Random forest	24.05 ± 0.86
LSTM (raw score)	21.87 ± 1.01

Table 2 compares score separability. Table 3 isolates the agent at a threshold estimated without test data. The agent is not assigned an additional post-hoc EER; its actual causal decisions are reported.

Table 3. Frozen-threshold and EMA-agent operating points in the balanced alternating stream (mean ± standard deviation over five folds).

$\|\delta\|_2$	Decision rule	FAR (%)	FRR (%)	HTER (%)
0	Frozen	22.55 ± 1.13	23.82 ± 2.27	23.19 ± 0.95
0	EMA agent	24.46 ± 2.12	24.60 ± 2.04	24.53 ± 1.28
1	Frozen	22.43 ± 0.69	24.82 ± 1.97	23.63 ± 1.10
1	EMA agent	24.30 ± 2.00	25.73 ± 2.64	25.02 ± 1.66
2	Frozen	22.40 ± 0.85	27.37 ± 2.91	24.89 ± 1.37
2	EMA agent	24.53 ± 2.11	28.25 ± 4.25	26.39 ± 2.12

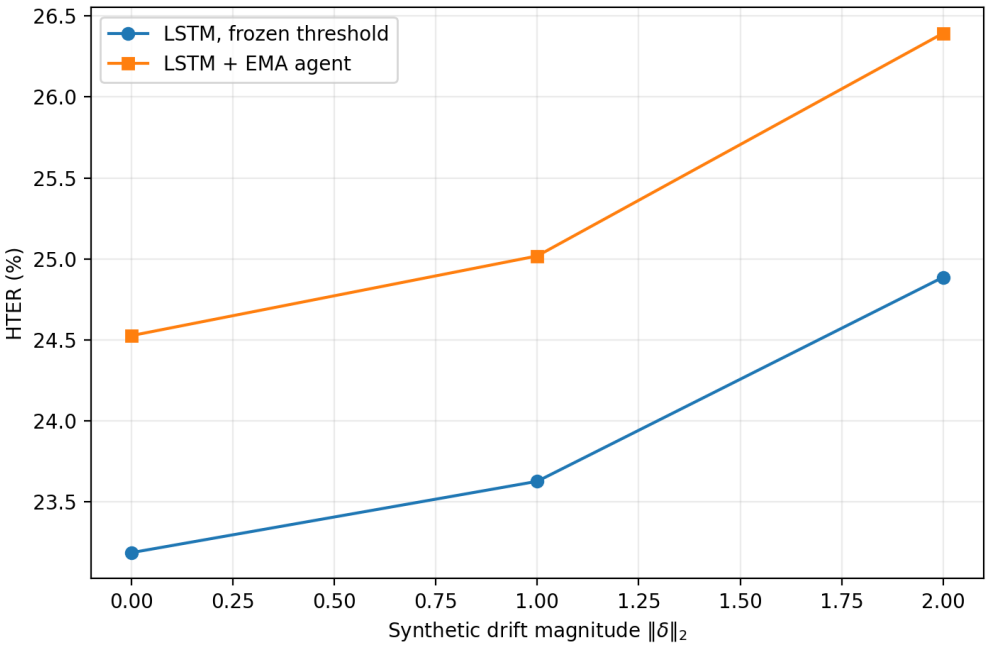


Fig. 1. HTER of the frozen LSTM threshold and EMA agent under synthetic standardized-feature drift. Error bars are omitted because fold-level values are reported in the accompanying result file.

On clean data, the LSTM lowers macro EER by 2.18 percentage points relative to RF and by 8.05 points relative to the linear SVM. These values are much higher than the invalid figures produced when metadata columns entered the feature vector, illustrating why the leakage correction is substantive.

The agent ablation is negative. On clean sessions, adaptation increases HTER by 1.34 points. At drift magnitudes 1 and 2, it increases HTER by 1.39 and 1.51 points, respectively. The frozen threshold has lower HTER in every one of the five folds at all three drift levels. Adaptation increases both FAR and FRR on average; therefore its poorer HTER cannot be characterized as a deliberate security–usability trade. Both decision rules degrade as drift grows, and the tested EMA mechanism does not preserve the operating point better than the static comparator.

7. Discussion

The revised experiment separates three questions that were previously mixed. First, Table 2 measures whether the LSTM score distinguishes a claimant from zero-effort impostors. Second, Table 3 measures whether an independently initialized operating point is usable without recalibration. Third, Fig. 1 asks whether EMA adaptation changes that operating point under a controlled shift. A low EER alone does not prove that the agent contributes, and a lower FRR is not a benefit if FAR rises enough to worsen HTER.

The accepted-only update is intentionally lightweight and causal. It avoids the clearly unsafe behavior of lowering a threshold after rejection, but it cannot adapt to scores that fall below the current threshold. It also remains exposed to poisoning after false acceptance. A deployable design should combine high-confidence update gates, step-up authentication, and explicit poisoning tests.

7.1. Limitations

Only HMOG is evaluated. This directly limits external validity and does not satisfy the stronger claim that the method generalizes across devices, populations, or sensor corpora. A second session-rich dataset is required before making such a claim. Six empty sessions are skipped and documented. The modulo-session split prevents sample overlap but is not a rolling-origin forecast of a newly enrolled user. The alternating 50% impostor stream is a security stress test rather than a real-world prevalence model. The synthetic drift is defined in feature space and uses one direction per fold; future work should include multiple directions, sensor bias/scale/rotation perturbations, natural late-session drift, confidence intervals clustered by session, and live on-device latency and energy measurements.

Comparisons with published numbers are intentionally not ranked. Prior studies use different activities, modalities, enrollment sizes, subject sets, and split rules, so claiming superiority from EER alone would be misleading.

8. Conclusion

This work presents a reproducible evaluation of an LSTM with an EMA-updated authentication threshold on HMOG. The revised pipeline removes identifier leakage, uses complete session-disjoint partitions, separates validation from training, provides the missing LSTM-only ablation, and tests the adaptive mechanism under controlled drift. The LSTM has the best clean EER among the three classifiers, but the EMA agent consistently worsens HTER relative to the same LSTM with a frozen threshold. The tested agent should therefore not be deployed or presented as a successful drift solution. The evidence supports only the measured HMOG protocol; broader effectiveness requires independent-dataset and live-stream validation.

Reproducibility Statement

The companion code records sensor-column selection, fold assignments, skipped sessions, random seed, hyperparameters, per-fold results, and aggregate summaries. The canonical entry point is `hmog_revised_experiment.py`; legacy result files are excluded from the revised manuscript.

References

[1] A. Alzubaidi, J. Kalita, Authentication of smartphone users using behavioral biometrics, IEEE Communications Surveys & Tutorials 18 (3) (2016) 1998–2026. doi:10.1109/COMST.2016.2537748.

[2] I. Stylios, S. Kokolakis, O. Thanou, S. Chatzis, Behavioral biometrics & continuous user authentication on mobile devices: A survey, Information Fusion 66 (2021) 76–99.

[3] Z. Sitová, J. Šeděnka, Q. Yang, G. Peng, G. Zhou, P. Gasti, K. S. Balagani, HMOG: New behavioral biometric features for continuous authentication of smartphone users, IEEE Transactions on Information Forensics and Security 11 (5) (2016) 877–892.

[4] M. Abuhamad, A. Abusnaina, D. Nyang, D. Mohaisen, Sensor-based continuous authentication of smartphones' users using behavioral biometrics: A contemporary survey, IEEE Internet of Things Journal 8 (1) (2021) 65–84. doi:10.1109/JIOT.2020.3020076.

[5] A. Verma, V. Moghaddam, A. Anwar, Data-driven behavioural biometrics for continuous and adaptive user verification using smartphone and smartwatch, Sustainability 14 (12) (2022) 7362. doi:10.3390/su14127362.

[6] L. Gonzalez-Manzano, J. M. De Fuentes, A. Ribagorda, Leveraging user-related internet of things for continuous authentication: A survey, ACM Computing Surveys 52 (3) (2019) 53:1–53:38. doi:10.1145/3314023.

[7] S. Ayeswarya, J. Norman, A survey on different continuous authentication systems, International Journal of Biometrics 11 (1) (2019) 67–99. doi:10.1504/IJBM.2019.096574.

[8] Y. Li, H. Hu, G. Zhou, Using data augmentation in continuous authentication on smartphones, IEEE Internet of Things Journal 6 (1) (2019) 628–640.

[9] M. Abuhamad, T. Abuhmed, D. Mohaisen, D. Nyang, AUToSen: Deep-learning-based implicit continuous authentication using smartphone sensors, IEEE Internet of Things Journal 7 (6) (2020) 5008–5020.

[10] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Computation 9 (8) (1997) 1735–1780. doi:10.1162/neco.1997.9.8.1735.

[11] X. Zeng, X. Zhang, S. Yang, Z. Shi, C. Chi, Gait-based implicit authentication using edge computing and deep learning for mobile devices, Sensors 21 (13) (2021) 4592. doi:10.3390/s21134592.

[12] G. Giorgi, A. Saracino, F. Martinelli, Using recurrent neural networks for continuous authentication through gait analysis, Pattern Recognition Letters 147 (2021) 157–163. doi:10.1016/j.patrec.2021.03.010.

[13] G. J. Ross, N. M. Adams, D. K. Tasoulis, D. J. Hand, Exponentially weighted moving average charts for detecting concept drift, Pattern Recognition Letters 33 (2) (2012) 191–198. doi:10.1016/j.patrec.2011.08.019.

[14] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, A. Bouchachia, A survey on concept drift adaptation, ACM Computing Surveys 46 (4) (2014) 44. doi:10.1145/2523813.